

Using temporal orientation, category fluency, and word recall for detecting cognitive impairment: the 10-point cognitive screener (10-CS)

Daniel Apolinario¹, Daniel Gomes Lichtenthaler¹, Regina Miksian Magaldi¹, Aline Thomaz Soares¹, Alexandre Leopold Busse¹, Jose Renato das Gracas Amaral¹, Wilson Jacob-Filho¹ and Sonia Maria Dozzi Brucki²

¹Division of Geriatrics, Department of Internal Medicine, Faculdade de Medicina, Universidade de São Paulo, São Paulo, Brazil

²Division of Behavioral and Cognitive Neurology, Department of Neurology, Faculdade de Medicina, Universidade de São Paulo, São Paulo, Brazil

Correspondence to: Daniel Apolinario, E-mail: daniel.apolinario@usp.br

Objectives: A screening strategy composed of three-item temporal orientation and three-word recall has been increasingly used for detecting cognitive impairment. However, the intervening task administered between presentation and recall has varied. We evaluated six brief tasks that could be useful as intervening distractors and possibly provide incremental accuracy: serial subtraction, clock drawing, category fluency, letter fluency, timed visual detection, and digits backwards.

Methods: Older adults ($n=230$) consecutively referred for suspected cognitive impairment underwent a comprehensive assessment for gold-standard diagnosis, of whom 56 (24%) presented cognitive impairment not dementia and 68 (30%) presented dementia. Among those with dementia, 87% presented very mild or mild stages (Clinical Dementia Rating 0.5 or 1). The incremental value of each candidate intervening task in a model already containing orientation and word recall was assessed.

Results: Category fluency (animal naming) presented the highest incremental value among the six candidate intervening tasks. Reclassification analyses revealed a net gain of 12% among cognitively impaired and 17% among normal participants. A four-point scaled score of the animal naming task was added to three-item temporal orientation and three-word recall to compose the 10-point Cognitive Screener. The education-adjusted 10-point Cognitive Screener outperformed the longer Mini-Mental State Examination for detecting both cognitive impairment (area under the curve 0.85 vs 0.77; $p=0.027$) and dementia (area under the curve 0.90 vs 0.83; $p=0.015$).

Conclusions: Based on empirical data, we have developed a brief and easy-to-use screening strategy with higher accuracy and some practical advantages compared with commonly used tools. Copyright © 2015 John Wiley & Sons, Ltd.

Key words: orientation; memory; verbal fluency; dementia; mild cognitive impairment; screening

History: Received 9 October 2014; Accepted 17 February 2015; Published online in Wiley Online Library (wileyonlinelibrary.com)

DOI: 10.1002/gps.4282

Introduction

Dementia is highly unrecognized in medical settings, where between 42% and 67% of the affected individuals are not properly diagnosed (O'Connor *et al.*, 1988; Jacinto *et al.*, 2011). Rates of unrecognized dementia are even higher in early stages. Physicians fail

to make diagnosis in up to 91% of the patients with mild dementia (Valcour *et al.*, 2000).

Subjective factors affect the recognition of cognitive impairment and previous studies have reported that objective tests can identify impaired subjects more accurately than physicians in regular clinical encounters (Borson *et al.*, 2006). Accordingly, recent

guidelines suggest that early detection of cognitive impairment should not rely on informal observation alone, but mainly in structured screening tools (Cordell *et al.*, 2013).

Structured assessments of mental status and tests of global cognitive functioning almost invariably include measures of temporal orientation and episodic memory. Item analyses of the Mini-Mental State Examination (MMSE) have demonstrated that, among all items, word recall and temporal orientation are those with the best properties for detecting dementia (Galasko *et al.*, 1990; Fillenbaum *et al.*, 1994).

The six-item screener (SIS) is composed of three-item recall (apple, table, and penny) and three-item temporal orientation (day, month, and year). Each correct response earns one point for a total of six points. The express rationale is to combine the properties of two simple subtests for detecting dementia, with word recall presenting better sensitivity and orientation presenting better specificity. In the validation study conducted with two independent cohorts, the SIS performed comparably with the longer MMSE (Callahan *et al.*, 2002). Subsequent studies confirmed acceptable properties of the SIS for detecting cognitive impairment (Wilber *et al.*, 2005; Chen *et al.*, 2010).

In studies using the SIS, the intervening task administered between presentation of words and free recall has varied. On occasions where the SIS score was calculated based on existing information from MMSE items (Callahan *et al.*, 2002; Chen *et al.*, 2010), the intervening task was evidently the same as what is used in the MMSE (i.e., serial sevens or spelling “world” backwards). Wilber *et al.* (2005) have used only the orientation subtest as an intervening task. Carpenter *et al.* (2011) reported that recall was asked after 1 min, but did not mention the task administered during the delay period.

In a consecutive sample of older adults referred for suspected cognitive impairment, we searched for a brief task that could be useful as an intervening distractor, and, at the same time, would provide increased accuracy when combined with the SIS, eventually originating a new screening tool.

Methods

Participants and diagnostic procedures

Older patients referred for suspected cognitive impairment were recruited prospectively in a government-financed outpatient geriatric clinic in the city of São

Paulo, Brazil. The eligibility criteria included age ≥ 60 years, suspicion of cognitive impairment as the main reason for referral, and the availability of a knowledgeable informant to be interviewed face-to-face or by telephone. We did not include patients with moderate to severe dementia, as indicated by dependence for basic activities of daily living, because in these cases the diagnosis is evident and cognitive tests may be otiose. We have also excluded individuals with delirium and those who had sensory, motor, or speech disturbances that precluded completion of the neuropsychological assessment.

Participants were interviewed for demographic information, including age, gender, race (White/non-White), and education (highest grade completed). Economic status was determined according to the Brazilian Economic Classification Criterion, which provides a continuous scale calculated by assigning scores to the number of household assets (Brazilian Association of Research Companies, 2008).

Participants underwent a comprehensive protocol that included a 60-min neuropsychological assessment composed by the Mini-Mental State Examination (MMSE; Folstein *et al.*, 1975), the Sunderland Clock Drawing Test (Sunderland *et al.*, 1989), the Free and Cued Selective Reminding Test (Grober and Buschke, 1987), and the NEUROPSI neuropsychological test battery (Ostrosky-Solís *et al.*, 1999). The NEUROPSI battery consists of 26 subtests that were developed to assess a variety of cognitive functions and were culturally appropriate for Latin American populations (Abrisqueta-Gomez *et al.*, 2008). In all cases, the Clinical Dementia Rating (CDR) interview was administered to a knowledgeable informant (Morris, 1993). The CDR sum of boxes score (CDR-SB) has been adopted as an overall measure to rate the severity of the cognitive impairment, with scores ranging from 0 to 18 (O’Bryant *et al.*, 2010).

Dementia was diagnosed according to the Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (American Psychiatric Association, 1994). A diagnosis of Cognitive Impairment No Dementia (CIND) was made when the subject did not fulfill standard criteria for dementia, but presented impairment in one or more domains, as described by Ebly *et al.* (1995). In this study, impairment in a given domain was operationalized as score ≤ 1.5 standard deviations (SD) below normative means in at least one subtest or ≤ 1 SD in at least two subtests representative of the following domains: attention, memory, language, visuo-constructive functions, and executive functions. Scores were interpreted using the best available education-adjusted norms judged adequate for

the sociocultural background of the sample (Ostrosky-Solís *et al.*, 1999; Brucki and Rocha, 2004; Grober *et al.*, 2008; Lourenço *et al.*, 2008). The final diagnosis for each participant was reached at a consensus conference with the presence of at least two experts. Based on all the available data, participants were assigned to one of the following mutually exclusive diagnostic categories: normal cognitive aging, CIND, or dementia. In analyses of binary outcomes, relevant diagnoses were defined as dementia and cognitive impairment (CIND + dementia).

An informed consent was obtained before the interview. The next of kin consented on behalf of the participants judged by the interviewer as having questionable capacity to understand the study procedures. The protocol was approved by the local research ethics committee.

Six-item screener and candidate intervening tasks

In this study, a slightly modified version of the SIS has been used. We have chosen “date” in place of “day of week”, because prior studies have shown that date is a more discriminating item (Ashford *et al.*, 1989; Solfrizzi *et al.*, 2001). Calculation of the SIS score was based on existing information from MMSE items by summing the three-item recall of the Brazilian version (Brucki *et al.*, 2003) and three of the temporal orientation items (date, month, and year).

Six tasks were evaluated that can be administered in about 1 min and could serve as candidate intervening distractors between learning and delayed recall: (1) NEUROPSI Digits Backwards. A random list of numbers is read out and individuals repeat the numbers in the reverse order. Sequences start with two digits and increase progressively up to six digits; (2) NEUROPSI Visual Detection. The subject is requested to cross out figures in a sheet containing 16 targets and 240 distractors in 60 s; (3) Serial Threes Subtraction. Beginning with the number 20, participants subtract three from the previous result in five sequential steps; (4) Animal Naming. Subjects are asked to generate as many names of animals as possible in 60 s; (5) Letter Fluency. Subjects generate as many words beginning with the letter “F” as possible in 60 s; and (6) Sunderland Clock Drawing. Subjects are asked to draw the face of a clock and set the time to 2:45. These measures were taken in a single section not as actual intervening distractors, but rather as isolated tasks presented in the same order as listed earlier.

Statistical analysis

The incremental value of each candidate intervening task was examined using the integrated discrimination improvement (IDI) as described by Pencina *et al.* (2008). The IDI, calculated by comparing discrimination slopes of models with binary outcomes, assesses the improvement in accuracy obtained by adding a new piece of information to a set of existing predictors. The selected intervening task was further examined using the net reclassification improvement (NRI), a measure of upward and downward movement of among risk categories that offers an intuitive way of quantifying the improvement in predicted probabilities offered by a new measure. NRI has been calculated as a sum of two separate components: one for cognitively impaired individuals and the other for individuals with normal cognitive aging. The proportion of subjects moving accurately or inaccurately from one quartile of risk to another was calculated after adding the intervening task into the model. IDI and NRI are relatively new methods for assessing improvement in model performance that have been considered to be more powerful and less influenced by the strength of the baseline model when compared with previously used measures (Pencina *et al.*, 2012).

Ordinal logistic regression models were fitted to examine the independent ability of each subtest in predicting the cognitive status. In these models, the dependent variable was formed with the three ordered diagnostic categories (normal, CIND, and dementia), the subtests were entered simultaneously as independent variables, and the unstandardized beta coefficients were reported. The relative contribution of each subtest in predicting the cognitive status was further examined through a multiple linear regression model taking the CDR-SB as the dependent variable.

Receiver operating characteristic (ROC) curves were obtained for each cognitive measure and the area under the curve (AUC) was reported with its 95% confidence interval (95% CI). In ROC curve analyses, the discriminative power of each measure was evaluated by contrasting cognitively impaired patients (CIND + dementia) with normal patients and demented patients with non-demented patients (normal + CIND). The nonparametric methods described by DeLong *et al.* (1988) were used for calculating standard errors and assessing differences between AUCs. The Youden index (sensitivity + specificity – 1) was computed for each cutoff to determine the thresholds with optimal discriminative performance.

Statistical analyses were performed with Stata version 13.1 (Stata Corp. LP, College Station, TX, USA).

All statistical tests were two-tailed, and an alpha level of less than 0.05 was used to indicate statistical significance.

Results

From June 2009 to February 2011, 1306 new patients consecutively referred to a geriatric consultation were screened for participation. Among these, 380 were referred for suspected cognitive impairment and thus were considered for inclusion. We have excluded 79 individuals who needed assistance with basic activities of daily living, 45 who failed to come for the neuropsychological testing appointment, 9 for refusing participation, 8 for lacking an informant, 8 for presenting significant sensory, motor, or speech disturbances, and 1 for delirium. Excluded individuals were older (77.6 vs 74.7 years; $p < 0.001$) and marginally more likely to be women (74.3% vs 64.3%; $p = 0.052$) when compared with participants.

Our final sample consisted of 230 older adults with a mean (SD) age of 74.7 (7.2) years, 64.3% women, and an average 4.7 (4.2) years of schooling. After a standardized evaluation, 106 (46.1%) of the patients were diagnosed with normal cognitive aging, 56 (24.3%) with CIND, and 68 (29.6%) with dementia. Among those with CIND, 92.9% had CDR=0.5, and the remaining had CDR 0. Among those with dementia, 44.1% had CDR 0.5, 42.6% had CDR=1, and the remaining had CDR=2. Demographic and clinical data of participants are presented in Table 1.

Among the six candidate intervening tasks evaluated in this study, five significantly improved the accuracy of the SIS to detect cognitive impairment, but only two (animal naming and visual detection) have been shown to improve the accuracy of the SIS to detect dementia (Table 2).

The animal naming test has been chosen as the intervening task to be used between learning and recall, as it presented the highest IDI for both detecting

Table 1 Characteristics of the study participants according to the diagnostic group ($n = 230$)

	Normal ($n = 106$)	CIND ($n = 56$)	Dementia ($n = 68$)	p -value*
Demographic characteristics				
Age (years), median (IQR)	72 (67–78)	77 (71–79)	79 (73–84)	<0.001
Gender (female), n (%)	82 (77.4)	33 (58.9)	33 (48.5)	<0.001
Race (White), n (%)	75 (70.8)	33 (58.9)	48 (70.6)	0.261
Education (years), median (IQR)	4 (2–8)	4 (2–4)	4 (1–5)	0.091
Economic Status — BECC, median (IQR)	22 (16–27)	19 (16–23)	21 (17–27)	0.084
Clinical characteristics				
Charlson comorbidity index, median (IQR)	1 (0–2)	1 (0–2)	1 (1–2)	0.404
Prescription drugs in use, median (IQR)	6 (3–8)	6 (5–7)	5 (2–8)	0.738
Major depression, n (%)	27 (25.5)	17 (30.4)	13 (19.1)	0.345
Gait disturbance, n (%)	7 (6.6)	11 (19.6)	31 (45.6)	<0.001
Clinical Dementia Rating (CDR)				
CDR 0.0, n (%)	106 (100)	4 (7.1)	—	<0.001
CDR 0.5, n (%)	—	52 (92.9)	30 (44.1)	
CDR 1.0, n (%)	—	—	29 (42.6)	
CDR 2.0, n (%)	—	—	9 (13.2)	
Sum of boxes, median (IQR)	0 (0–0)	2 (1–2)	5 (4–7)	<0.001
Cognitive measures				
Mini-Mental State Examination, median (IQR)	26 (23–28)	24 (21–27)	20 (16–22)	<0.001
Serial subtraction, median (IQR)	5 (4–5)	4 (4–5)	4 (2–5)	0.002
Clock drawing, median (IQR)	9 (4–9)	5 (4–9)	4 (4–5)	<0.001
Animal naming, median (IQR)	13 (11–16)	11 (8–14)	7 (3–10)	<0.001
Letter fluency, median (IQR)	8 (5–12)	6 (3–9)	4 (1–7)	<0.001
Visual detection, median (IQR)	8 (6–11)	7 (3–9)	3 (2–5)	<0.001
Digits backwards, median (IQR)	3 (3–4)	3 (2–4)	3 (2–3)	<0.001

CIND, cognitive impairment no dementia; IQR, interquartile range; BECC, Brazilian Economic Classification Criterion.

Data are shown as the median (interquartile range) or number (percentage).

*The Kruskal–Wallis test has been used to compare interval data between the diagnostic groups and the chi-squared test has been used for categorical data.

Table 2 Accuracy of the potential intervening tasks as isolated measures and integrated discrimination improvement in addition to the orientation and memory

	Cognitive impairment (CIND + dementia)				Dementia			
	AUC (95% CI)	IDI	SE	<i>p</i> -value	AUC (95% CI)	IDI	SE	<i>p</i> -value
Serial subtraction	0.61 [0.54, 0.68]	0.00	0.00	0.962	0.62 [0.55, 0.70]	0.00	0.00	0.911
Clock drawing	0.68 [0.60, 0.76]	0.03	0.01	0.006	0.72 [0.64, 0.80]	0.00	0.01	0.960
Animal naming	0.80 [0.75, 0.86]	0.08	0.02	<0.001	0.83 [0.78, 0.89]	0.05	0.02	0.016
Letter fluency	0.72 [0.65, 0.78]	0.05	0.01	0.001	0.72 [0.64, 0.79]	0.01	0.01	0.136
Visual detection	0.77 [0.71, 0.83]	0.06	0.02	<0.001	0.80 [0.74, 0.86]	0.03	0.01	0.034
Digits backwards	0.67 [0.61, 0.74]	0.02	0.01	0.018	0.67 [0.60, 0.75]	0.01	0.01	0.347

AUC, area under receiver operating characteristic curves; CIND, cognitive impairment no dementia; IDI, integrated discrimination improvement; SE, standard error.

cognitive impairment and dementia. A scaled score with five levels was developed based on quintiles, with approximations intended to make it easier for the examiners to remember the scoring levels when administering the test from memory. We verified that the scaling procedure did not significantly reduce the AUC of the animal naming test for detection of cognitive impairment (0.80 vs 0.79; $p=0.159$) or dementia (0.83 vs 0.82; $p=0.123$).

Net reclassification after adding the animal naming task into a model already containing orientation and memory revealed significant improvement, with an overall NRI was estimated as 0.30 ($z=3.61$; $p<0.001$). Among the 124 cognitively impaired participants, 30 (24.2%) were correctly reclassified into a higher risk category, whereas 15 (12.1%) were inappropriately reclassified into a lower risk category, with a net gain of 12.1%. Among the 106 cognitively normal participants, 32 (30.2%) were correctly reclassified into a lower risk category, whereas 14 (13.2%) were incorrectly reclassified into a higher risk category, with a net gain of 17.0%. In an ordinal logistic regression model, orientation to time ($\beta=-1.15$; $p<0.001$), delayed recall ($\beta=-0.72$; $p<0.001$), and animal naming ($\beta=-0.70$; $p<0.001$) were independent predictors of the cognitive status. In a multiple linear regression model taking the CDR-SB as the dependent variable, orientation to time ($\beta=-1.58$; $p<0.001$), delayed recall ($\beta=-0.44$; $p=0.001$), and animal naming ($\beta=-0.61$; $p<0.001$) remained as significant predictors and, when combined, accounted for 55% of the variance.

Therefore, a scaled score of the animal naming test (0–4 points) was added to the temporal orientation items (0–3 points) and the delayed recall items (0–3 points) to compose a test that we have named the 10-point Cognitive Screener (10-CS). Instructions for administration and scoring of the 10-CS are presented in Figure 1.

A positive correlation between 10-CS scores and years of education was observed in participants with normal cognitive aging ($r=0.35$; $p<0.001$). To adjust the score for education effects, 2 points were added for unschooled individuals and one point for those with 1–3 years of schooling. Adjustment was unnecessary for individuals with at least 4 years of schooling, because education effects were not significant above that level. After adjusting the 10-CS, a sum of up to 12 points would be theoretically possible, but we have limited the score to a maximum of 10 points. The resulting measure was called the education-adjusted 10-CS (10-CS-Edu). This procedure was intended to improve the stability of sensitivity and specificity across educational levels. As expected, the adjustment did not significantly change the accuracies for detecting cognitive impairment (AUC 0.85 vs 0.85; $p=0.745$) or dementia (AUC 0.91 vs 0.90; $p=0.607$).

In a multiple linear regression model conducted with the subsample of normal participants, 10-CS-Edu was not associated with age ($\beta=0.00$; $p=0.960$), gender ($\beta=0.30$; $p=0.378$), education ($\beta=-0.01$; $p=0.770$), race ($\beta=0.35$; $p=0.280$), or economic status ($\beta=0.02$; $p=0.422$). These variables predicted only 3% of the variance, indicating that 10-CS-Edu is not significantly affected by demographic characteristics.

The education-adjusted cutoff with the highest Youden index was 6/7 points for both cognitive impairment and dementia. At that cutoff, 10-CS-Edu presented sensitivity of 60.5% and specificity of 94.3% for cognitive impairment, with an AUC of 0.85 (95% CI [0.79, 0.89]). For detecting dementia, the 10-CS-Edu presented sensitivity of 80.9% and specificity of 84.0%, with an AUC of 0.90 (95% CI [0.86, 0.94]). Sensitivity, specificity, and likelihood ratios for each cutoff point of the 10-CS-Edu are shown in Table 3.

Although the Youden index is a useful measure to assess the trade-off between sensitivity and specificity,

10-point Cognitive Screener (10-CS)	
▶ WHAT YEAR IS THIS?	0 1
▶ WHAT MONTH IS THIS?	0 1
▶ WHAT IS TODAY'S DATE?	0 1
NOW I AM GOING TO NAME THREE OBJECTS. PLEASE WAIT UNTIL I SAY ALL THREE WORDS, THEN REPEAT THEM. REMEMBER WHAT THEY ARE. I AM GOING TO ASK YOU TO NAME THEM AGAIN IN A FEW MINUTES. PLEASE REPEAT THESE WORDS FOR ME: CAR—VASE—BRICK <i>(may repeat 3 times if necessary; repetition is not scored)</i>	
▶ NOW TELL ME THE NAMES OF ALL ANIMALS YOU CAN THINK OF, AS QUICK AS POSSIBLE. I'LL GIVE YOU ONE MINUTE TO COME UP WITH AS MANY ANIMALS AS YOU CAN. READY?	
0 - 5:	0
6 - 8:	1
9 - 11:	2
12 - 14:	3
15 or more:	4
▶ NOW WHAT WERE THE THREE OBJECTS I ASKED YOU TO REMEMBER?	
CAR	0 1
VASE	0 1
BRICK	0 1
10-CS : _____	
Adjustment for Education Effects: ▫ No formal education: add 2 points (to a max of 10) ▫ 1-3 years of education: add 1 point (to a max of 10)	
10-CS-Edu : _____	
Interpretation of the 10-CS-Edu: ▫ ≥ 8 points: Normal ▫ 6-7 points: Possible cognitive impairment ▫ 0-5 points: Probable cognitive impairment	

Figure 1 Instructions for administration and scoring of the 10-point Cognitive Screener.

the cutoff with the highest index may not be the best option in some clinical settings. As can be seen in Table 3, the cutoff 6/7 would provide low sensitivity for detecting milder stages of cognitive impairment. Therefore, we have developed a decision rule for screening in clinical practice. Individuals with 10-CS-Edu ≥ 8 ($n=101$; 43.9% of our sample) can be assumed to have normal or near-normal cognitive performance. Among these subjects, 73.3% were

normal, 22.8% presented CIND, and 4.0% presented dementia. Individuals with a corrected score ≤ 5 ($n=57$; 24.8% of our sample) present probable cognitive impairment. In this subsample, 82.5% had dementia, 14.0% had CIND, and only 3.5% were normal. Individuals with a corrected score of six or seven ($n=72$; 31.3% of our sample) present possible cognitive impairment and should undergo a more detailed assessment. These subjects represent a mixed group with 41.7% of normal cognitive aging, 34.7% of CIND, and 23.6% of dementia.

When compared with the MMSE, the 10-CS-Edu presented accuracies that were significantly higher both for cognitive impairment (AUC 0.85 vs 0.77; $p=0.027$) and for dementia (AUC 0.90 vs 0.83; $p=0.015$). Figure 2 illustrates the comparative accuracies of the screening tools with ROC curve plots.

In a sensitivity analysis, we estimated what would have been the performance of the 10-CS-Edu using day of the week instead of date. There was a non-significant trend towards a lower accuracy for day of the week as compared with date, with accuracies of 77.3 vs 81.2% ($p=0.253$) for dementia and 58.5 vs 64.2% ($p=0.087$) for cognitive impairment. The discriminative power of the 10-CS-Edu was comparable with day of the week and date both for dementia (AUC 0.90 vs 0.90; $p=0.894$) and cognitive impairment (AUC 0.85 vs 0.85; $p=0.741$). Accordingly, it is possible to assume that date and day of the week could be used interchangeably without substantial changes on the properties of the instrument.

Discussion

In a consecutive sample of older adults with suspected cognitive impairment, we have identified two brief tasks that, besides being useful as 1-min intervening distractors, can improve the properties of a pre-existing instrument composed of three-item recall and three-item temporal orientation. Therefore, we propose a screening tool with a significantly higher accuracy at the cost of a negligible increase in complexity and time for completion.

Animal naming and visual detection were the tasks able to improve the accuracy of the SIS for detecting both cognitive impairment and dementia. Although these measures assess considerably different aspects of cognition, both are timed tasks that involve processing speed and impose demands upon executive control processes, abilities that decline early in dementia (Carlson *et al.*, 2009). Moreover, those tasks provide complementary information about one's

Table 3 Sensitivity, specificity, and likelihood ratios for the education-adjusted scores of the 10-point Cognitive Screener according to each cutoff point

Cutoff	Cognitive impairment (CIND + dementia)				Dementia			
	Sensitivity (95% CI)	Specificity (95% CI)	+LR	−LR	Sensitivity (95% CI)	Specificity (95% CI)	+LR	−LR
≤1	3.23 [0.9, 8.1]	100 [96.6, 100]		0.97	5.88 [1.6, 14.4]	100 [97.7, 100]		0.94
≤2	9.68 [5.1, 16.3]	100 [96.6, 100]		0.90	17.7 [9.5, 28.8]	100 [97.7, 100]		0.82
≤3	21.0 [14.2, 29.2]	100 [96.6, 100]		0.79	33.8 [22.8, 46.3]	98.2 [94.7, 99.6]	18.3	0.67
≤4	35.5 [27.1, 44.6]	100 [96.6, 100]		0.65	58.8 [46.2, 70.6]	97.5 [93.8, 99.3]	23.8	0.42
≤5	44.4 [35.4, 53.5]	98.1 [93.4, 99.8]	23.5	0.57	69.1 [56.7, 79.8]	93.8 [88.9, 97.0]	11.2	0.33
≤6	60.5 [51.3, 69.1]	94.3 [88.1, 97.9]	10.7	0.42	80.9 [69.5, 89.4]	84.0 [77.4, 89.2]	5.04	0.23
≤7	78.2 [69.9, 85.1]	69.8 [60.1, 78.3]	2.59	0.31	94.1 [85.6, 98.4]	59.9 [51.9, 67.5]	2.35	0.10
≤8	91.1 [84.7, 95.5]	53.8 [43.8, 63.5]	1.97	0.16	97.1 [89.8, 99.6]	40.7 [33.1, 48.7]	1.64	0.07
≤9	95.2 [89.8, 98.2]	22.6 [15.1, 31.8]	1.23	0.21	100 [94.7, 100]	18.5 [12.9, 25.4]	1.23	

CIND, cognitive impairment no dementia; +LR, positive likelihood ratio; −LR, negative likelihood ratio.

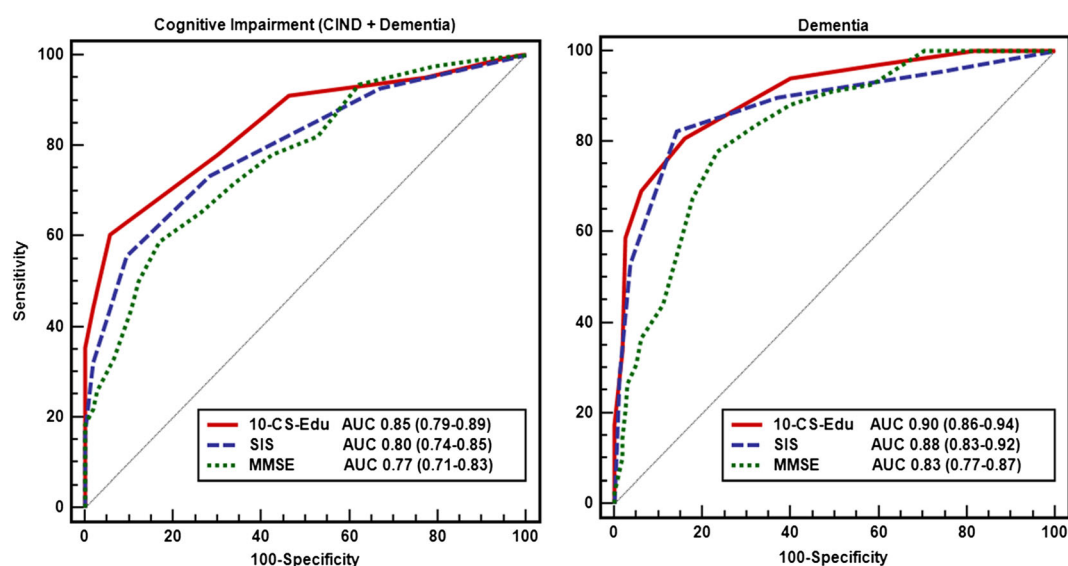


Figure 2 The ROC curve plots illustrating comparative accuracies of 10-CS-Edu, SIS, and MMSE. Sensitivity and specificity of the cognitive measures are presented in the form of ROC curves. Reported values indicate area under ROC curve and 95% confidence interval. ROC, receiver operating characteristic; AUC, area under the curve; CIND, cognitive impairment no dementia; 10-CS-Edu, education-adjusted 10-point Cognitive Screener; SIS, Six-Item Screener; MMSE, Mini-Mental State Examination.

cognitive profile in addition to orientation and memory.

One of the most widely used cognitive screeners, the Mini-Cog, is composed of a three-word recall and a clock drawing task (Borson *et al.*, 2000). In our study, the clock drawing test did not provide useful information for detecting dementia in addition to orientation and recall. In fact, recent studies have demonstrated that the Mini-Cog performs poorly in heterogeneous, predominantly low-educated populations (Filho and Lourenço, 2009; Carnero-Pardo

et al., 2013). Therefore, in populations with such characteristics, the 10-CS seems to be an advantageous alternative.

A particular concern may arise when using fluency tasks as intervening distractors. Some authors suggest that, in tests involving learning and recall of words, a non-verbal task should be administered during the delay interval in order to minimize effects of retroactive interference on recall of target words (Delis *et al.*, 2000). However, there is little evidence to suggest that such concern is necessary. Indeed, a recent study

demonstrated that the administration of verbal intervening tasks in older adults did not elicit more retro-active interference than non-verbal intervening tasks during the delay interval of the California Verbal Learning Test (Willians *et al.*, 2014).

Some strengths of this study should be highlighted. First, we have gathered a naturalistic sample of consecutive patients referred for with suspected cognitive impairment, thus ensuring good external validity to our findings. Second, gold-standard diagnoses were obtained by consensus of experts after a comprehensive assessment, including the CDR interview with a knowledgeable informant in all cases. Third, a range of possible intervening distractor tasks with differing characteristics have been evaluated for composing the 10-CS.

Our study has some limitations that warrant consideration. First, 10-CS was not administered in its final format to patients. Scores were rather calculated using available information of the corresponding subtests. Second, investigators who established the diagnosis were not blind to the information used for deriving 10-CS scores, raising concerns regarding diagnostic circularity. Third, while the 10-CS presented an adequate performance in this low-educated sample, it is not clear whether a possible ceiling effect would affect its sensitivity in populations with higher educational levels.

Besides having higher accuracy than MMSE, 10-CS presents important features that make it particularly advantageous for use in clinical practice: (1) it takes only about 2 min to complete; (2) it does not require specific materials; (3) it has an easy-to-remember structure to be administered and scored from memory; (4) it is adequate to low-educated populations for not requiring reading, writing, drawing, and other abilities gained through schooling; (5) it does not include visual cues or motor tasks and thus can be completed without considerable disadvantage by subjects with motor or sensory deficiencies that are common in older age; and (6) it can be easily administered by telephone.

Conclusion

Based on empirical data, our study demonstrated that a combination of subtests assessing orientation, word recall, and verbal fluency presents encouraging performance for screening dementia and an acceptable performance for screening any cognitive impairment. Therefore, we propose an easy-to-administer screening tool with potential advantages when compared

with longer tests such as the MMSE. A further formal validation should be carried out to confirm the properties of the 10-CS in non-specialist settings.

Conflict of interest

None declared.

Key points

- A four-point scaled score of the animal naming task was added to three-item temporal orientation and three-word recall to compose the 10-point Cognitive Screener (10-CS).
- The education-adjusted 10-CS outperformed the longer Mini-Mental State Examination for detecting both cognitive impairment and dementia.
- Based on empirical data, we have developed a brief and easy-to-use screening strategy with higher accuracy and some practical advantages compared to commonly used tools.

References

- Abrisqueta-Gomez J, Ostrosky-Solis F, Bertolucci PH, Bueno OF. 2008. Applicability of the abbreviated neuropsychologic battery (NEUROPSI) in Alzheimer disease patients. *Alzheimer Dis Assoc Disord* 22: 72–78.
- American Psychiatric Association. 1994. *Diagnostic and Statistical Manual of Mental Disorders*. 4th edn. Washington, DC: American Psychiatric Association Press.
- Ashford JW, Kolm P, Colliver JA, Bekian C, Hsu LN. 1989. Alzheimer patient evaluation and the mini-mental state: item characteristic curve analysis. *J Gerontol* 44: P139–146.
- Borson S, Scanlan J, Brush M, Vitaliano P, Dokmak A. 2000. The mini-cog: a cognitive “vital signs” measure for dementia screening in multi-lingual elderly. *Int J Geriatr Psychiatry* 15: 1021–1027.
- Borson S, Scanlan JM, Watanabe J, Tu SP, Lessig M. 2006. Improving identification of cognitive impairment in primary care. *Int J Geriatr Psychiatry* 21: 349–355.
- Brazilian Association of Research Companies. 2008. Brazilian Economic Classification Criterion. Available: <http://www.abep.org/criterioBrasil.aspx>. Accessed: 12 February 2015.
- Brucki SM, Nitrini R, Caramelli P, Bertolucci PH, Okamoto IH. 2003. Suggestions for utilization of the Mini-Mental State Examination in Brazil. *Arq Neuropsiquiatr* 61: 777–781.
- Brucki SM, Rocha MS. 2004. Category fluency test: effects of age, gender and education on total scores, clustering and switching in Brazilian Portuguese-speaking subjects. *Braz J Med Biol Res* 37: 1771–1777.
- Callahan CM, Unverzagt FW, Hui SL, Perkins AJ, Hendrie HC. 2002. Six-item screener to identify cognitive impairment among potential subjects for clinical research. *Med Care* 40: 771–781.
- Carlson MC, Xue QL, Zhou J, Fried LP. 2009. Executive decline and dysfunction precedes declines in memory: the Women’s Health and Aging Study II. *J Gerontol A Biol Sci Med Sci* 64: 110–117.
- Carnero-Pardo C, Cruz-Orduña I, Espejo-Martínez B, *et al.* 2013. Utility of the mini-cog for detection of cognitive impairment in primary care: data from two Spanish studies. *Int J Alzheimers Dis* 2013: 285462.
- Carpenter CR, DesPain B, Keeling TN, Shah M, Rothenberger M. 2011. The Six-Item Screener and AD8 for the detection of cognitive impairment in geriatric emergency department patients. *Ann Emerg Med* 57: 653–661.
- Chen MR, Guo QH, Cao XY, Hong Z, Liu XH. 2010. A preliminary study of the Six-Item Screener in detecting cognitive impairment. *Neurosci Bull* 26: 317–321.

- Cordell CB, Borson S, Boustani M, *et al.* 2013. Alzheimer's Association recommendations for operationalizing the detection of cognitive impairment during the Medicare Annual Wellness Visit in a primary care setting. *Alzheimers Dement* **9**: 141–150.
- Delis DC, Kramer J, Kaplan E, Ober BA. 2000. *The California Verbal Learning Test-2*. The Psychological Corporation: San Antonio, TX.
- DeLong ER, DeLong DM, Clarke-Pearson DL. 1988. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics* **44**: 837–845.
- Ebly EM, Hogan DB, Parhad IM. 1995. Cognitive impairment in the nondemented elderly. Results from the Canadian Study of Health and Aging. *Arch Neurol* **52**: 612–619.
- Filho STR, Lourenço RA. 2009. The performance of the Mini-Cog in a sample of low educational level elderly. *Dement Neuropsychol* **3**: 81–87.
- Fillenbaum GG, Wilkinson WE, Welsh KA, Mohs RC. 1994. Discrimination between stages of Alzheimer's disease with subsets of Mini-Mental State Examination items: an analysis of Consortium to Establish a Registry for Alzheimer's Disease data. *Arch Neurol* **51**: 916–921.
- Folstein MF, Folstein SE, McHugh PR. 1975. Mini-mental state. A practical method for grading the cognitive state of patients for the clinician. *J Psychiatr Res* **12**: 189–198.
- Galasko D, Klauber MR, Hofstetter CR, *et al.* 1990. The Mini-Mental State Examination in the early diagnosis of Alzheimer's disease. *Arch Neurol* **47**: 49–52.
- Grober E, Buschke H. 1987. Genuine memory deficits in dementia. *Dev Neuropsychol* **3**: 13–36.
- Grober E, Hall C, McGinn M, *et al.* 2008. Neuropsychological strategies for detecting early dementia. *J Int Neuropsychol Soc* **14**: 130–142.
- Jacinto AF, Brucki S, Porto CS, Martins Mde A, Nitrini R. 2011. Detection of cognitive impairment in the elderly by general internists in Brazil. *Clinics (Sao Paulo)* **66**: 1379–1384.
- Lourenço RA, Ribeiro-Filho ST, Moreira Ide F, Paradelo EM, Miranda AS. 2008. The Clock Drawing Test: performance among elderly with low educational level. *Rev Bras Psiquiatr* **30**: 309–315.
- Morris JC. 1993. The Clinical Dementia Rating (CDR): current version and scoring rules. *Neurology* **43**: 2412–2414.
- O'Bryant SE, Lacritz LH, Hall J, *et al.* 2010. Validation of the new interpretive guidelines for the clinical dementia rating scale sum of boxes score in the national Alzheimer's coordinating center database. *Arch Neurol* **67**: 746–749.
- O'Connor DW, Pollitt PA, Hyde JB, *et al.* 1988. Do general practitioners miss dementia in elderly patients? *BMJ* **297**: 1107–1110.
- Ostrosky-Solis F, Ardila A, Rosselli M. 1999. NEUROPSI: a brief neuropsychological test battery in Spanish with norms by age and educational level. *J Int Neuropsychol Soc* **5**: 413–433.
- Pencina MJ, D'Agostino RB Sr, D'Agostino RB Jr, Vasan RS. 2008. Evaluating the added predictive ability of a new marker: from area under the ROC curve to reclassification and beyond. *Stat Med* **27**: 157–172.
- Pencina MJ, D'Agostino RB, Pencina KM, Janssens AC, Greenland P. 2012. Interpreting incremental value of markers added to risk prediction models. *Am J Epidemiol* **176**: 473–481.
- Solfrizzi V, Torres F, Capurso C, *et al.* 2001. Analysis of individual items of Mini-Mental State Examination in discrimination between normal and demented subjects. *Arch Gerontol Geriatr* **7**: 357–362.
- Sunderland T, Hill JL, Mellow AM, *et al.* 1989. Clock drawing in Alzheimer's disease. A novel measure of dementia severity. *J Am Geriatr Soc* **37**: 725–729.
- Valcour VG, Masaki KH, Curb JD, Blanchette PL. 2000. The detection of dementia in the primary care setting. *Arch Intern Med* **160**: 2964–2968.
- Wilber ST, Lofgren SD, Mager TG, Blanda M, Gerson LW. 2005. An evaluation of two screening tools for cognitive impairment in older emergency department patients. *Acad Emerg Med* **12**: 612–616.
- Williams BR, Sullivan SK, Morra LF, Williams JR, Donovan PJ. 2014. The similar effects of verbal and non-verbal intervening tasks on word recall in an elderly population. *Clin Neuropsychol* **28**: 505–513.